Chapter 3
Analyzing Geometric Integrators

In the previous chapter, we discussed the growth of error in numerical methods for differential equations. We saw that if the time interval is fixed, the error obeys the power law relationship with stepsize that is predicted by the convergence theory. We also saw that this did not contradict the exponential growth in the error with time (when the stepsize is fixed). The latter issue casts doubt on the reliance on the convergence order as a means for assessing the suitability of an integrator for molecular dynamics.

At the same time, tantalizingly, the Verlet method appears to exhibit a remarkable energy conservation property that was not predicted by the standard convergence theorem, even as the error itself grows exponentially fast. This is particularly surprising as the Verlet method does not seem to be constructed in a fundamentally different way to the Euler method or other schemes such as explicit Runge-Kutta methods (which do not exhibit a similar conservation property). That is, they use force evaluations at a few points in the step to approximate the solution at the next step. What is it that Verlet has that many other methods lack? A strong hint has been given in the previous section, where we learned that the Verlet method is a symplectic method, but it is still not obvious why this should lead to a qualitatively different energy conservation behavior.

In this chapter, we show that a symplectic integrator can be viewed as being effectively equivalent to the flow map of a certain Hamiltonian system. The starting point is that symplectic integrators are symplectic maps that are "near to the identity" since they depend on a parameter (the stepsize h) which can be chosen as small as needed, and, if consistent, in the limit $h \to 0$, such a map must tend to the identity map. We can express the fundamental consequence as follows: *not only are Hamiltonian flow maps symplectic, but also near-identity symplectic maps are (in an approximate sense) Hamiltonian flow maps [31].* The fact leads to the existence of a modified (perturbed) Hamiltonian from which the discrete trajectory may be derived (as snapshots of continuous trajectories). In some cases we may derive this perturbed Hamiltonian as an expansion in powers of the stepsize.

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B. Leimkuhler, C. Matthews, *Molecular Dynamics*, Interdisciplinary Applied Mathematics 39, DOI 10.1007/978-3-319-16375-8_3